

413/613 Fall 2026 - Applied Statistics & Data Science I - Lecture 4

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RECALL

From experiments to probability distributions

$$\text{experiment} \longrightarrow \Omega \longrightarrow X(\omega) \longrightarrow P(X = x).$$

A random variable assigns a numerical value to each outcome of an experiment. Its probability distribution tells us how likely the different values of X are. For a discrete random variable,

$$E[X] = \sum_x x P(X = x), \quad \text{Var}(X) = \mathbb{E}[(X - \mu)^2] = E[X^2] - \mu^2.$$

The expectation describes the *center* of the distribution, while the variance describes its *spread* around that center.

For repeated independent Bernoulli trials,

$$H \sim \text{Binomial}(N, p), \quad P(H = h) = \binom{N}{h} p^h (1 - p)^{N-h},$$

with

$$\mathbb{E}[H] = Np, \quad \text{Var}(H) = Np(1 - p).$$

We also saw how different limiting regimes lead to different distributions:

$$\text{Binomial} \xrightarrow[N \rightarrow \infty, p \rightarrow 0]{Np = \lambda} \text{Poisson}(\lambda)$$

for rare events, while for large N ,

$$\text{Binomial}(N, p) \approx \mathcal{N}(Np, Np(1 - p)).$$

1 From Probability to Statistics

So far, we have mostly studied *probability*: given a probability distribution, what can we say about the observations generated from it? Statistics asks the reverse question. We observe only a finite collection of samples and would like to learn something about the unknown population from which they were generated:

Probability: Model $\longrightarrow$ Data, Statistics: Data $\longrightarrow$ Model.

Before developing statistical methods, we first need to understand a fundamental question:

Why should a finite collection of random observations tell us anything reliable about the underlying population?

The Law of Large Numbers will provide our first answer. We begin with two simple probability inequalities.

2 Markov's Inequality

Let X be a nonnegative random variable, $X \geq 0$. For any $a > 0$,

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

This gives an upper bound on the probability that a nonnegative random variable takes a large value, using only its expectation. To see why, write

$$\mathbb{E}[X] = \int_0^\infty x p(x) dx.$$

Since the integrand is nonnegative,

$$\mathbb{E}[X] \geq \int_a^\infty x p(x) dx.$$

But $x \geq a$ on this interval, so

$$\mathbb{E}[X] \geq a \int_a^\infty p(x) dx = aP(X \geq a).$$

Therefore,

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Markov's inequality is very general: we do not need to know the full distribution of X . NOTE: We did the derivation of this inequality using discrete RV in the class.

2.1 Example: ICU admissions

Suppose X denotes the number of patients admitted to emergency on a given day. Assume that the average number of admissions is

$$\mathbb{E}[X] = 5.$$

With no other knowledge about the distribution of X , what can we say about the probability of observing at least 20 admissions?

Since $X \geq 0$, Markov's inequality gives

$$P(X \geq 20) \leq \frac{\mathbb{E}[X]}{20} = \frac{5}{20} = 0.25$$

Therefore,

$$P(X \geq 20) \leq 0.25.$$

Notice what this statement does *not* say. **It does not say that the probability is 0.25.** The actual probability might be 0.1, 0.05, or even smaller. Markov's inequality only guarantees that it cannot exceed 0.25.

This is useful because we obtained a probability bound knowing only one number: the mean $\mathbb{E}[X]$.

3 Chebyshev's Inequality

Suppose X has mean and variance

$$\mathbb{E}[X] = \mu, \quad \text{Var}(X) = \sigma^2.$$

We would like to ask:

How likely is X to be far away from its mean?

Consider the nonnegative random variable

$$Y = (X - \mu)^2.$$

Applying Markov's inequality,

$$P(Y \geq \varepsilon^2) \leq \frac{\mathbb{E}[Y]}{\varepsilon^2}.$$

Since

$$\mathbb{E}[Y] = \mathbb{E}[(X - \mu)^2] = \sigma^2,$$

and

$$(X - \mu)^2 \geq \varepsilon^2 \iff |X - \mu| \geq \varepsilon,$$

we obtain

$$\boxed{P(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}.$$

Equivalently, writing $\varepsilon = k\sigma$,

$$\boxed{P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

The important point is that this statement does not assume that X is Gaussian. It holds for *any* distribution with finite variance.

3.1 Example: ICU admissions revisited

Suppose we now know not only the average number of ICU admissions,

$$\mathbb{E}[X] = 5,$$

but also the variance,

$$\text{Var}(X) = 4, \quad \sigma = 2.$$

What can we say about the probability that the number of admissions differs from the mean by at least 4 patients? We want

$$P(|X - 5| \geq 4).$$

Since $4 = 2\sigma$, Chebyshev's inequality gives

$$P(|X - 5| \geq 4) \leq \frac{\sigma^2}{4^2} = \frac{4}{16} = \frac{1}{4}.$$

Therefore,

$$P(|X - 5| \geq 4) \leq 0.25.$$

Equivalently,

$$P(|X - 5| < 4) \geq 0.75.$$

Thus, without assuming that the number of ICU admissions follows a Gaussian, Poisson, or any other particular distribution, we know that at least 75% of the probability must lie within 4 of the mean.

3.2 Markov versus Chebyshev

It is important to understand that the Markov and Chebyshev answer related but different questions.

Markov inequality: How likely is X to be large?

For a nonnegative random variable,

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

It requires only the mean $\mathbb{E}[X]$.

Chebyshev inequality instead asks

Chebyshev inequality: How likely is X to be far from its mean?

If X has mean μ and variance σ^2 ,

$$P(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}.$$

Thus,

	Information needed	Question
Markov	$\mathbb{E}[X]$	X unusually large?
Chebyshev	$\mathbb{E}[X], \text{Var}(X)$	X unusually far from its mean?

In fact, in the derivation of Chebyshev inequality follows directly from Markov inequality by applying Markov's inequality to the nonnegative random variable

$$Y = (X - \mu)^2.$$

4 Random Sampling

We now make the transition from probability to statistics. Imagine an unknown population described by a probability distribution $p(x)$ with

$$\mathbb{E}[X] = \mu, \quad \text{Var}(X) = \sigma^2.$$

Usually we do not observe the entire population. Instead, we collect a finite random sample

$$X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} p.$$

The notation “iid” means that the observations are

1. **independent**: observing one sample does not affect the others;
2. **identically distributed**: every X_i is drawn from the same population distribution $p(x)$.

Thus,

$$\mathbb{E}[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2, \quad i = 1, \dots, n.$$

The population mean μ is generally unknown. From our observed samples, the most natural estimate is the **sample mean**

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

An important conceptual point is that $\bar{X}_n$ is itself a random variable. If we repeated the experiment and collected another n observations, we would generally obtain a different value of $\bar{X}_n$.

5 Expectation of the Sample Mean

What should we expect the sample mean to be? Using linearity of expectation,

$$\begin{aligned} E[\bar{X}_n] &= E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] \\ &= \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] \\ &= \frac{1}{n} \sum_{i=1}^n \mu \\ &= \mu. \end{aligned}$$

Thus,

$$E[\bar{X}_n] = \mu.$$

Averaging does not shift the expected value, so it is an *unbiased* estimate of the true mean μ . On average, the sample mean is centered exactly at the population mean.

6 Variance of the Sample Mean

The more interesting question is how much $\bar{X}_n$ fluctuates from one sample to another. Since the observations are independent,

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i).$$

Therefore,

$$\begin{aligned} \text{Var}(\bar{X}_n) &= \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) \\ &= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) \\ &= \frac{1}{n^2} n\sigma^2 \\ &= \boxed{\frac{\sigma^2}{n}}. \end{aligned}$$

Hence,

$$\boxed{\text{SD}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}}.$$

This is one of the most important scaling laws in statistics:

$$\boxed{\text{uncertainty in an average} \sim \frac{1}{\sqrt{n}}}.$$

As we collect more independent observations, the center remains fixed at μ , while the fluctuations of the sample mean become smaller.

6.1 Example: Election polling

Suppose there are $N = 1,000,000$ eligible voters in an election. We would like to know the fraction of voters who support Candidate A.

Let

$$X_i = \begin{cases} 1, & \text{if voter } i \text{ supports Candidate A,} \\ 0, & \text{otherwise.} \end{cases}$$

If we could ask every voter, the population proportion would be

$$p = \mathbb{E}[X].$$

But surveying all $N = 1,000,000$ voters is expensive and unnecessary. Instead, suppose we randomly sample $n = 1000$ voters ($n \ll N$):

$$X_1, X_2, \dots, X_{1000}.$$

The sample mean is

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Since each X_i is either 0 or 1, $\bar{X}_n$ is simply the fraction of sampled voters who support Candidate A. For example, if 530 of the 1000 sampled voters support Candidate A, then

$$\bar{X}_n = \frac{530}{1000} = 0.53.$$

Thus we would estimate the unknown population proportion p by

$$\boxed{\hat{p} = \bar{X}_n = 0.53}.$$

The central statistical question is:

Why should the opinion of 1000 randomly selected voters tell us anything about a population of 1,000,000 voters?

This is precisely the question addressed by random sampling and the Law of Large Numbers. For the voter example, the experiment has a binary outcome

$$X_i \sim \text{Bernoulli}(p),$$

we know

$$\mathbb{E}[X_i] = p, \quad \text{Var}(X_i) = p(1 - p).$$

Therefore,

$$E[\bar{X}_n] = p$$

and

$$\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n} = \frac{p(1 - p)}{n}.$$

Thus the poll estimate is centered around the true population proportion, while its uncertainty decreases as the sample size increases:

$$\text{SD}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}} = \sqrt{\frac{p(1 - p)}{n}}.$$

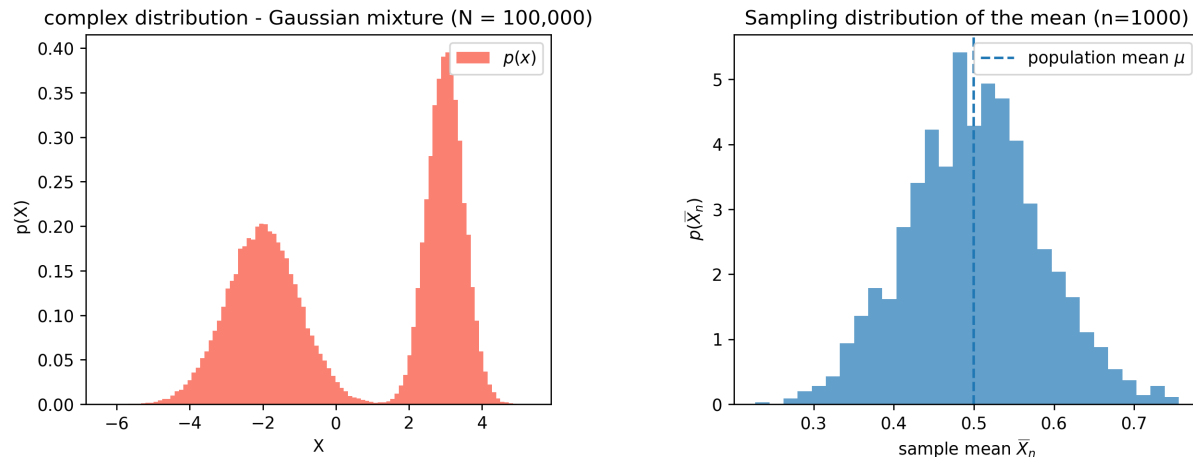


Figure 1: Population distribution and sampling distribution of the mean. Left: A complex population distribution $p(x)$, represented by $N = 100,000$ samples, with population mean μ and variance σ^2 . **Right:** We repeatedly draw $n = 1000$ observations from this population and compute the sample mean $\bar{X} = n^{-1} \sum_{i=1}^n X_i$. Repeating this experiment M times gives $\bar{X}^{(1)}, \dots, \bar{X}^{(M)}$; their histogram forms the empirical sampling distribution of the mean. The dashed line marks the population mean μ . Notice two striking features: the sample means are concentrated much more tightly around μ , and their distribution appears approximately Gaussian, despite the strongly non-Gaussian population distribution the samples were drawn from. We will return to this remarkable observation later.

Population, sample size, and repeated samples: Throughout the lecture, we distinguish between

- N = population size,
- n = number of observations in one random sample,
- M = number of times we repeat the sampling experiment.

Typically,

$$n \ll N.$$

Each experiment (random sampling) produces one sample mean

$$\bar{X}^{(m)} = \frac{1}{n} \sum_{i=1}^n X_i^{(m)}, \quad m = 1, \dots, M.$$

Repeating the experiment M times therefore produces

$$\bar{X}^{(1)}, \dots, \bar{X}^{(M)},$$

whose distribution is the empirical sampling distribution of the mean.

7 Law of Large Numbers

We have shown that

$$\mathbb{E}[\bar{X}_n] = \mu, \quad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

Applying Chebyshev's inequality to $\bar{X}_n$,

$$P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2}.$$

As $n \rightarrow \infty$,

$$\frac{\sigma^2}{n\varepsilon^2} \rightarrow 0.$$

Therefore,

$$\boxed{\bar{X}_n \xrightarrow{P} \mu.}$$

This is the **Law of Large Numbers**.

As we collect more independent observations, the sample mean concentrates around the population mean.

In the next lecture, we will discuss LLN and CLT in more detail.